STAFF REVIEW · H446 011 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

Do Now — source answers

1. A for loop . It is commonly used for count-controlled iteration.

2. && means AND . The whole expression is true only when both Boolean expressions are true.

3. 255 Eight bits give 256 patterns, representing values 0–255.

4. Sequence · iteration · branching/selection Instructions happen in order, a loop repeats code, and if chooses whether the pass counter changes.

I Do — source worked steps

1. Choose the return type — CODE:
public static bool IsEven(int number)
{
} — The result is true/false, so the return type is bool .

2. Use the remainder test — CODE:
public static bool IsEven(int number)
{
return number % 2 == 0;
} — The Boolean expression itself already produces true or false .

3. Use it from Main — CODE:
public static void Main()
{
bool result = IsEven(12);
Console.WriteLine(result);
} — The argument 12 is received by the parameter number .

4. Use contrasting values — CODE:
IsEven(12) → True
IsEven(7) → False
IsEven(0) → True

We Do — guided CHECK

1. Function or procedure?

2. Complete the method
CODE:
public static ? Square(int number)
{
?
}
CODE:

```
public static int Square(int number)
{
return number * number;
}
```

3. Design the module

4. Complete both statements

You Do — source mark schemes

1. DisplayTitle procedure
CODE:

```
using System;

public class Program
{
public static void Main()
{
DisplayTitle();
DisplayTitle();
}

public static void DisplayTitle()
{
Console.WriteLine("OCR H446 Computer Science");
}
}
```

Why it works: DisplayTitle() performs an action but does not return a value, so it uses void .

2. DoubleIt function
CODE:

```
using System;

public class Program
{
public static void Main()
{
int answer = DoubleIt(12);
Console.WriteLine(answer);
}

public static int DoubleIt(int number)
{
return number * 2;
}
}
```

Why it works: the method returns an integer, so the return type is int .

3. IsPass function
CODE:

```
using System;

public class Program
{
public static void Main()
{
Console.WriteLine(IsPass(65));
Console.WriteLine(IsPass(49));
}

public static bool IsPass(int mark)
{
return mark >= 50;
}
}
```

Why it works: the Boolean comparison already evaluates to true or false .

4. Refactor into modules
CODE:

```
using System;

public class Program
{
public static void Main()
{
Console.Write("First number: ");
int first = int.Parse(Console.ReadLine());

Console.Write("Second number: ");
int second = int.Parse(Console.ReadLine());

ShowLine();
int total = Add(first, second);
Console.WriteLine(total);
ShowLine();
}

public static int Add(int first, int second)
{
return first + second;
}

public static void ShowLine()
{
Console.WriteLine("----------");
}
}
```

Modularity: Add() has one calculation job, ShowLine() has one display job, and Main() coordinates them.

Exit Ticket — source answer

1. True. A function supplies a return value to the calling code.